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PlayList : spring boot full course
channel : india free internship
youtube channel https://www.youtube.com/@IndiaFreeInternship
source Code: [spring boot source Code]()
==
SESSION 1
----
1. SPRING FRAMEWORK
2. SPRING BOOT
3. SRPING DATA JPA
4. SPRING WEB MVC
5. SPRING REST
6. SPRING SECURITY
7. MICROSERVICE
8.SPRING BATCH
9.SPRING SCHEDULAR
==
SESSION 2
=========
Technolgies : webPage(servlets) +Database(JDBC/JPA) + Email (Java Mail) + Message(JMS)

Limitations:
- *Start coding from line zero*
- *Development takes more time*
- *More chance of error*
- *Design pattern applied manually*

Framework:
- *Ready-made structure*
- *In-built technologies*
- *Design patterns already implemented*

RAD => Rapid Application Development
- *Fast coding*
- *Less Error*

Spring Framework:
- *used for Web Application or Distributed appliation*
- *It has own Apis*
- *It Follow Container System*

Spring Container

- *Find/Scan Classes[spring bean]*
- *Create Object*
- *provide DATA*
- *Link Object based on Relation*
- *Finally Destroy*

Depedency Injection (DI)

Depedency Injection is a concept/theory which is implemented by spring IoC (Inversion of control) also called as spring container.

Theory	*programming*
oops	*core java*
ORM	*HIbernate with JPA*
DI	*Spring Ioc(Inversion of control)*

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session 3

=========

Application

====================

3 Type of application

(i) Web Application

(ii) Distributed Application

(iii) Microservices Application

(i) Web Application : Collection of web pages run under webserver.

Application : - n services

Project :- n Module

eg : Gmail Application

User(Register and Login) , Inbox, Sent, Drfats, Setting, ... etc

Amazon Application

User(Register and Login) , search, Cart, Payment, ...etc

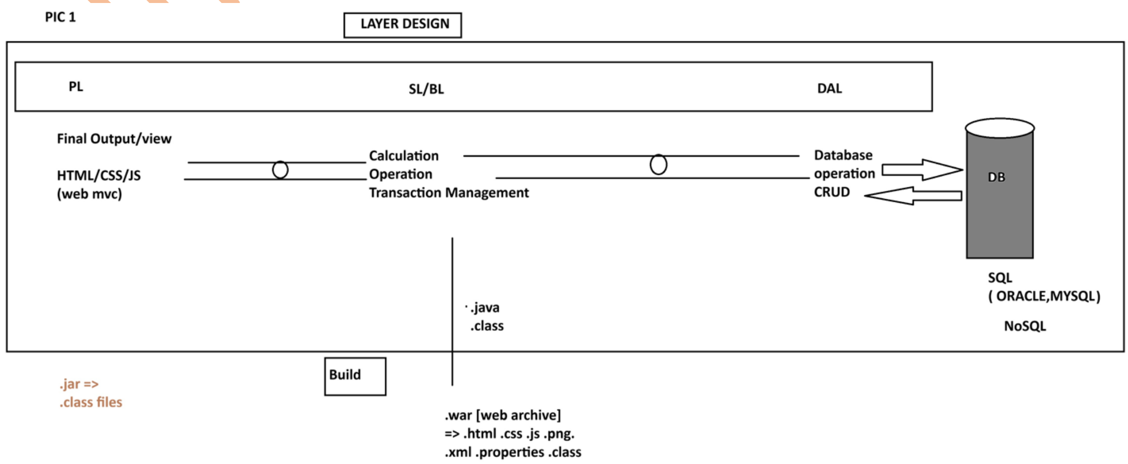
=> Implementation of service / module: - This is implemented using 3 Layers Design.

1. DAL : Data Access Layer (Database Operations)

2. SL : Service Layer (Calculation/Operations/txn mngmt)

3. PL : Presentation layer (view /Dynamic web pages/MVC)

PIC 1



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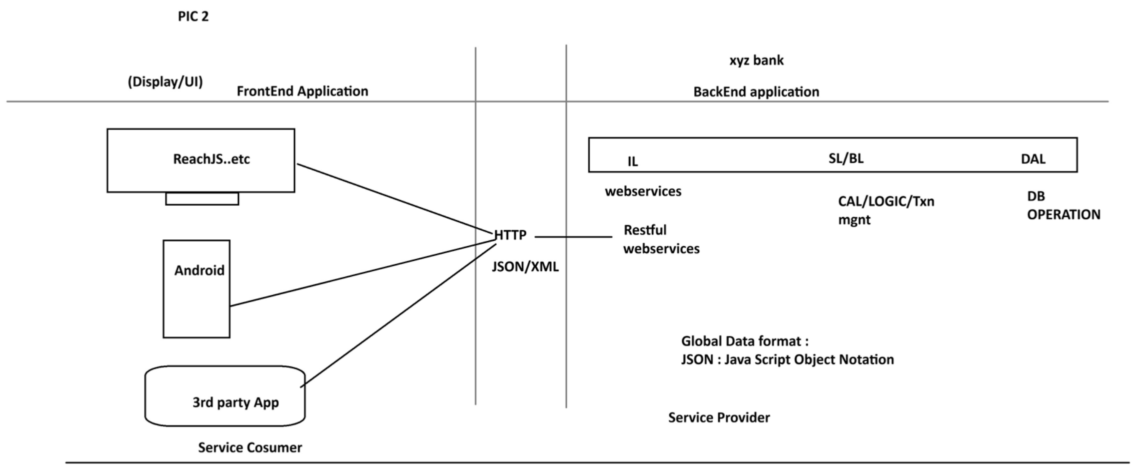
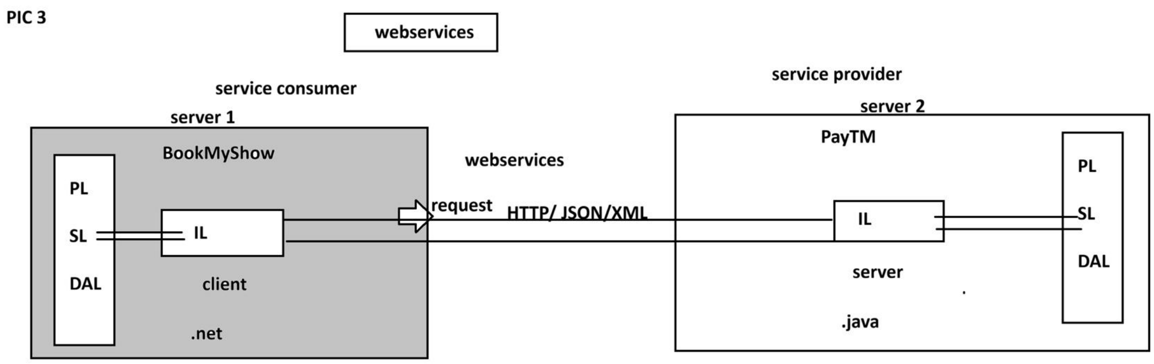
(ii) Distributed Application : An Application that runs at mutiple devices is called as distributed appliation.
xyz Bank application
This appliation can be accessed in different ways. Example.
eg:
1. Manually Process
2. NetBanking (Web App)
3. Mobile Banking
4. IVR
5. ATM / DC
6. 3rd party app (Gpay, PayPhone..etc)
.....

-But here business logics exist at backend application.
- Fontend applications are implemented using Angular, Android...etc
- App connected using HTTP Protocol using Global Data Format (JSON/XML)
PIC 2



WebServices:
===========
Intergration of multiple (at least two) applications which runs on different servers and implemented using different language (java, .net, php, python....etc)

PIC 3



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session 4

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SPRING FRAMEWORK
================
CORE
----
1. DI and IOC
2. Container Types
3. Annotation Configuration
4. @ComponentScan
5. @Component
6. @Value
7. Java Configuration
8. @Configuration
9. @Bean
10. @PropertySource
11. Environment
12. Lifecycle methods
13. Runners
14. @ConfigurationProperties
15. Properties file
16. YAML File
17. Profiles
18. Lombok API

===========================
spring core chapter 1
===============
Core:
This module/chapter gives all rules and guidelines to work with spring container.
1. DI and IOC
============

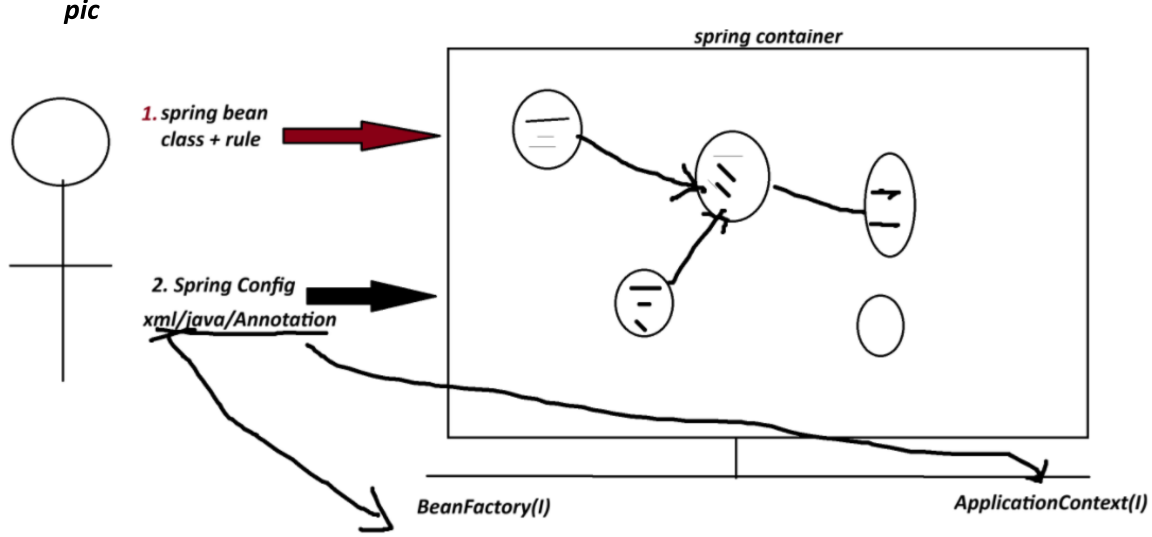
Spring Container :- It takes care of object.
1. Find/Scan our Classes(spring bean)
2. Create Object
3. Provide Data
4. Link Object
5. Destroy the Object

=>For this programmer has to provide two inputs-
1. Spring Bean : class+rules given by spring container.
2. Spring Configuration : This is main input cotains details like object name, relation with objects, data....etc.
xml/java/Annotation

*** Spring Container 2 Types:*
1. BeanFactory (Old container) : support only XML coonfiguration file.

2. ApplicationContext (New container) : support xml/java/Annotation configuration.

pic



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Dependeny Injection (DI)

Theory	programming
OOPs	java
ORM	HIbernate with JPA
DI	Spring IoC(Spring Container) -> Inversion of Control

=> Depedency :- An instance variable (Field) exist inside a class (spring bean)

1. Primitive Type Dependeny (PTD) :- If a variable is created using (8+1) datatype.
(byte, short, int, long, float, double, boolean, char +string)
(Their wrapper classess too)

2. Collection Type Depedency (CTD) :- If a variable created using(4) types:
(java.util.package)
(List,Map,Set and Properties)

3. Reference Type Depedency (RTD) :- If a variable is creatd using other class/interface

Injection(4)
======
=> It give/provide/assign existed provide data to Depedency(Inside object)

1. Setter Injection (SI) :- provide data to variable using its set method(setter).

2. Constructor Injection(CI): provide data creating object using parameterized constructor.

3. LookUpMethod Injection(LMI) :- Spring bean Scope
[When parent bean is sigleton and child bean is prototype)

4. Interface Injection(II) : - Not Exist in Spring F/W.

```
class Employee{
  int id;
  String name;
  double sal;
}
```

id=~~0~~ 10
name=~~null~~ Raj
sal=~~0.0~~ 1200.0

Employee e=new Employee()

```
class Product{
  int id; //PTD
  String name; //PTD
  Boolean exist; //PTD

  List<String> models; //CTD
  Map<String,Integer> data; //CTD

  Vendor vob; //RTD
  Service sob; //RTD
}
```

```
class Vendor{}

interface Service{}
```

SESSION 5

===========

1. Spring Bean

2. STS Installation

Spring Bean

=========

-It is class that follow rules given by Spring Container.

Rule

-----

a. class must be public Type.

b. Recomended to keep in a package (basePackage Rule)

ex :

com.app.service

com.app.controller

c. Fields(Variables) are option, if required recomended to keep private.

private Integer empId;

private String empName;

d. If variable present in class,

=>define default constructor with setter/getter

=>(or/and) Parameterized Constructor.

ex:

```
public class Employee{
private Integer id;
private String name;
public Empoyee(){
}
 public void setId(Integer id){
this.id=id;
  }

public Integer getId(){
  return id;
  }
}
```

e. We can override Object(c) methods:

toString() equals() hashCode()

[because non-private, non-final,non-static]

f. Inheritance Rule:

Allowed ->Only Spring API is allowed (ie spring F/W classes and interface)

not allowed-> EJB, Servlet

g. Annotation Rule:

Only we can add Spring API Annotation and Inetegration API Annotations

(Hibernate with JPA, Restful webServices annotations);

ex:

@Component, @Service, @Repository.. etc.

STS Installation

=============

step 1:

go to official website

spring.io

link : - https://spring.io/tools#eclipse

step 2 download STS according your OS

step 3

Rigth click on ZIP file

click on "Extract Here"

step 4

Open extracted folder

Run : SpringToolsForEclipse

Final STEP STS installed successfully.

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session 6

=========

1. Spring Bean

Simple Java class that follow rules given by container.

2. Spring Configuration File(XML)

It gives object

details (name, data,links with other objects ... etc).

3. Test class=> Just find, container created or not? object is creatd or not?

===

Spring core XML Tags

===

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

---

1. To create Object

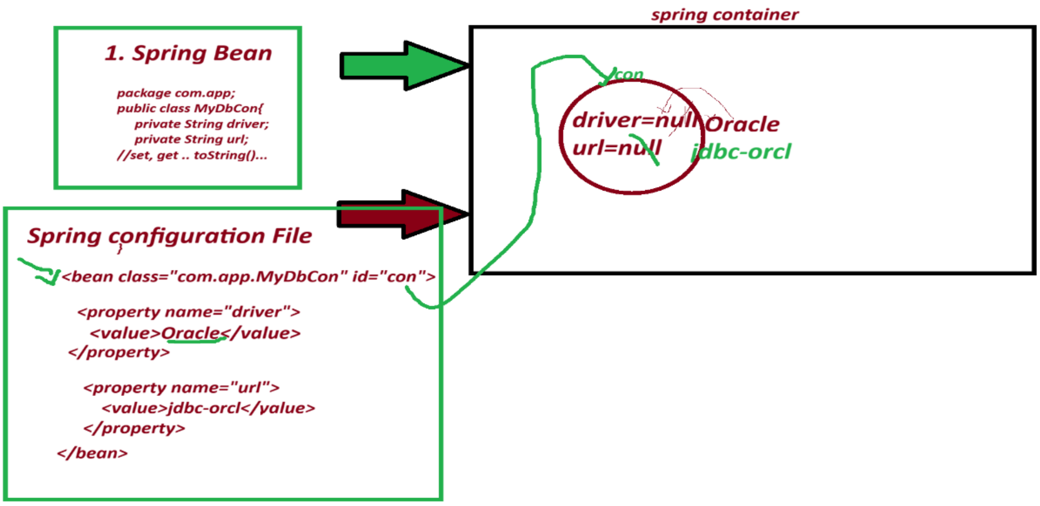
<bean id="objName" class="FullyQualifiedClassName"></bean>

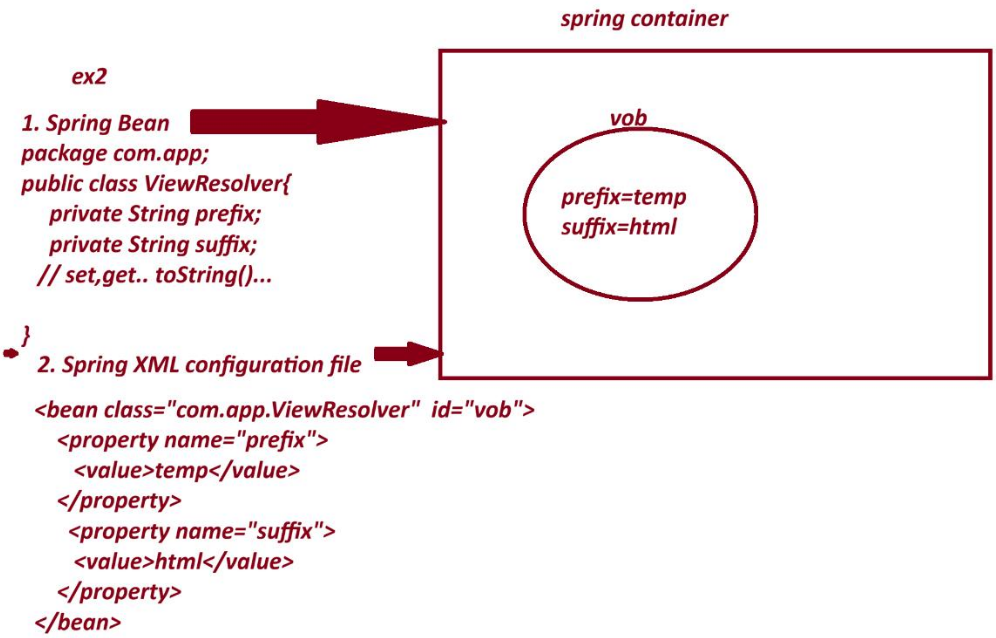
2. To call set method

<property name="variableName"></property>

3. To provide data to Primitive Variables

<value>data</value>





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1. Spring bean

```java
package com.app;

public class PaymentService{

    private int token;

    private boolean active;

    private String provider;

   //set.. set toString..etc

}
```

2. Spring XML configuration file.

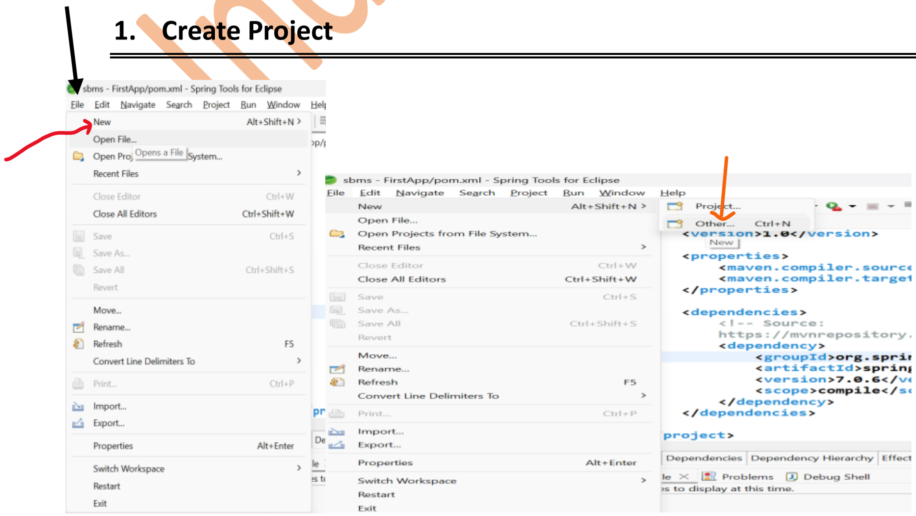
```xml
<bean class="com.app.PaymentService" id="psObj">

<property name="token">

<value>154678</value>

</property>

<property name="active">

<value>true</value>

</property>

<property name="provider">

<value>xyzBank</value>

</property>

</bean>
```

==

CREATE MAVEN PROJECT WITH BELOW STEP :-

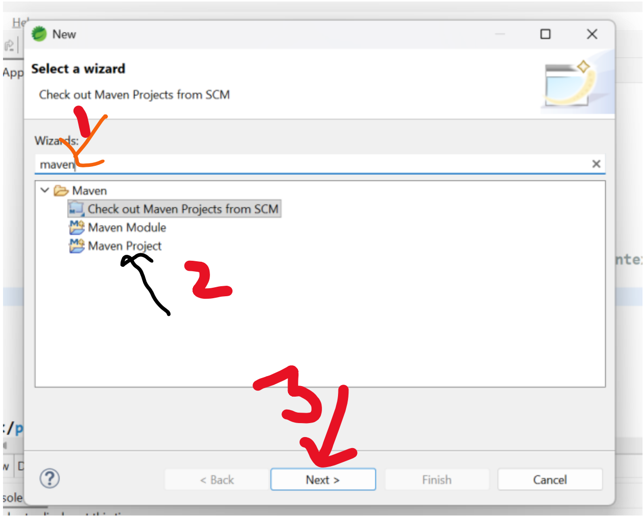
==

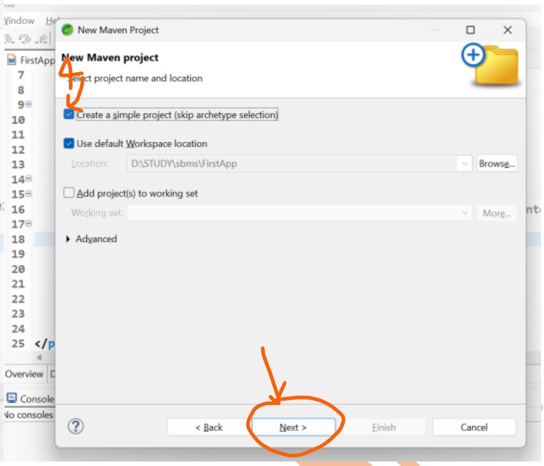
1. Create Project



File-> New-> Other-> Type maven->choose maven project-> next->

choose checkbox [v] create simple project -> next-> Enter details

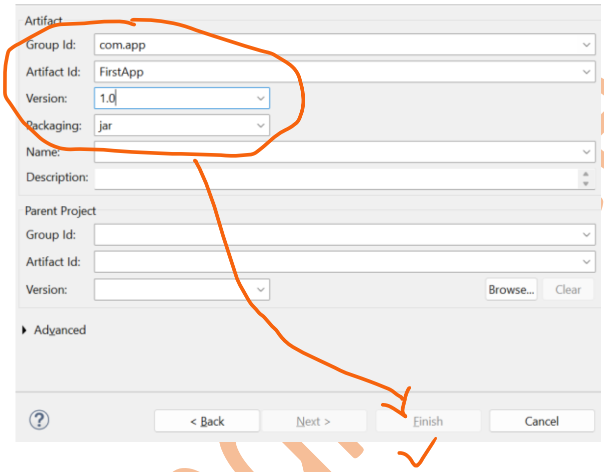




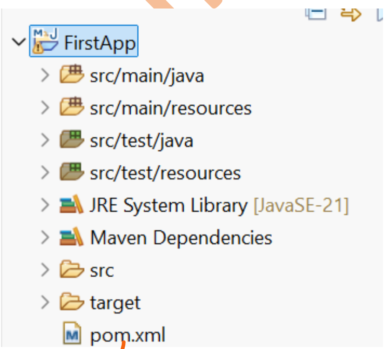
groupId : com.app

ArtficatId : FirstApp

version : 1.0



-> Finish



2. pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>FirstApp</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>

</properties>


<dependencies>

      <!-- Source:

        https://mvnrepository.com/artifact/org.springframework/spring-context -->

  <dependency>

    <groupId>org.springframework</groupId>

     <artifactId>spring-context</artifactId>

     <version>7.0.6</version>

     <scope>compile</scope>

  </dependency>

</dependencies>

</project>
```

---

session 7

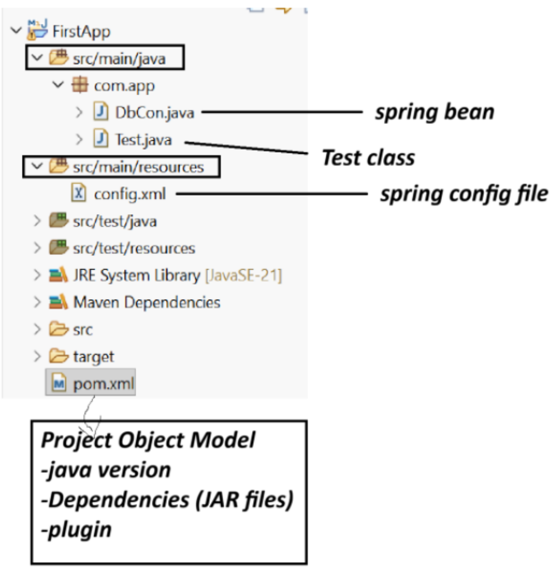
===========

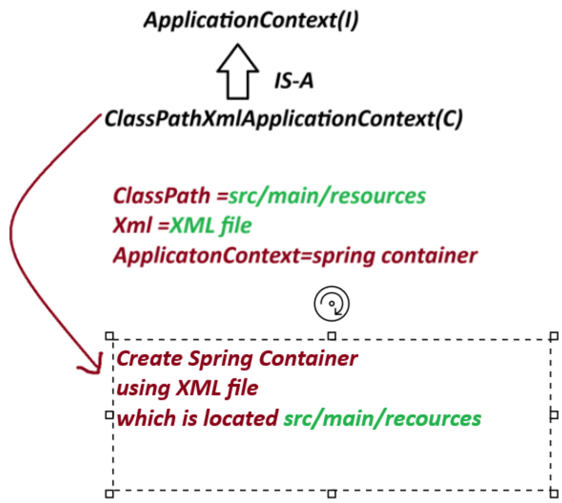
Spring Core First Application 3 Files.

1. Spring bean (.java)

2. Spring XML configuration (config.xml)

3. Test class(.java)





Note:-

1. Spring container are two Types.

(a) Legacy Container : BeanFactory(I) support only XML

(b) New Container : ApplicationContext(i) support all XML/java/Annotation config.

(c) ApplicationContext(I) : it is a interface.

we are using one of its Impl class:

ClassPathXmlApplicationContext(C).

===================================

Create Maven Project with below Step:

=================================

File>new>other>type maven>choose maven project>next>choose checkbox> create Simple project>next>Enter details

GroupId - com.app

ArtficatId - SecondApp

version - 1.0

----

code

SecondApp

=========

pom.xml

```
=============
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SecondApp</artifactId>
<version>1.0</version>

<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
---------------------------
```

DbCon.java

```
----------------------
package com.app;

public class DbCon {
  private String driver;
  private String url;
```

```
  //alt+shit+s
  public DbCon() {
super();
System.out.println("Constructor called");
  }
  public String getDriver() {
return driver;
  }
  public void setDriver(String driver) {
this.driver = driver;
System.out.println("Driver method called");
  }
  public String getUrl() {
return url;
  }
  public void setUrl(String url) {
this.url = url;
System.out.println("Url method called");
  }
  @Override
  public String toString() {
return "DbCon [driver=" + driver + ", url=" + url + "]";
  }

}
==========================================
```

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


    <bean class="com.app.DbCon" id="mobj">
        <property name="driver">
        <value>Oracle Driver</value>
        </property>


        <property name="url">
        <value>JDBC-ORCL</value>
        </property>
    </bean>


</beans>
```

=======================

Test.java

```
package com.app;


import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;


public class Test {
//ctrl+shit+O
public static void main(String[] args) {
ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
Object ob = ac.getBean("mobj");
System.out.println(ob);
}
}
```

----

output

Constructor called

Driver method called

Url method called

DbCon [driver=Oracle Driver, url=JDBC-ORCL]

---

session 8

=====================

Spring Container (DI - Depedency Injection)

=>Depedency - Variables exist in class

3 type

---

1. Primitive Type (8+1) : byte, short, int, long, float, double, boolean, char, String

2. Collection Type (4) : Set, List, Map and Properties

3. Reference Type (x) : using a class/interface variable

=> Injection : provide Data

- Setter Injection

- constructor Injection

-

LookUpMethod Injetion

- Field Injection (Annotations)

==

XML TAG

==

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

===

Collection Configuration

==

=> Spring container support working wiht collection configuration which are from java.util package

Collection Name	Collection Type	XML Tag Name
List	Interface	<list>
Set	Interface	<set>
Map	Interface	<map> and <entry>
Properties	Class	<props> and <prop>

List -> ArrayList

Set -> LinkedHasSet

Map -> LinkedHasMap

Properties --> NA

----

coding part collectionTypeS8

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>collectionTypeS8</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
```

```
</properties>

<dependencies><!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean

MyDetails.java

```
package com.app;

import java.util.List;
import java.util.Map;
import java.util.Properties;
import java.util.Set;

public class MyDetails {
//ctrl+shift+O
private List<String> data;
private Set<String> models;
private Map<Integer,String> design;
private Properties myinfo;

public MyDetails() {
super();
```

```
}

public List<String> getData() {
return data;
}

public void setData(List<String> data) {
this.data = data;
}

public Set<String> getModels() {
return models;
}

public void setModels(Set<String> models) {
this.models = models;
}

public Map<Integer, String> getDesign() {
return design;
}

public void setDesign(Map<Integer, String> design) {
this.design = design;
}
public Properties getMyinfo() {
return myinfo;
}public void setMyinfo(Properties myinfo) {
this.myinfo = myinfo;
}
@Override
public String toString() {
```

```java
return "MyDetails [data=" + data + ", models=" + models + ", design=" + design + ", myinfo=" + myinfo +
"]";

}

}
```

----

xml config file

config.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


<bean class="com.app.MyDetails" id="myinfo">

<property name="data">

<list>

<value>A</value>

<value>B</value>

<value>C</value>

<value>A</value>

</list>

</property>

<property name="models">

<set>

<value>M1</value>

<value>M2</value>

<value>M3</value>

<value>M1</value>

</set>

</property>
```

```xml
<property name="design">
<map>
<entry>
<key>
<value>101</value>
</key>
<value>AB</value>
</entry>
<entry>
<key>
<value>102</value>
</key>
<value>Suraj</value>
</entry>
<entry>
<key>
<value>103</value>
</key>

<null></null>
</entry>

</map>
</property>

<property name="myinfo">
<props>
<prop key="A1">B1</prop>
<prop key="A2">B2</prop>
<prop key="A3">B3</prop>
<prop key="A5"></prop>
</props>
```

```
</property>

</bean>

</beans>
```

Test File

Test.java

```java
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

public static void main(String[] args) {

ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
//Object ob = ac.getBean("myinfo");
MyDetails ob = ac.getBean("myinfo",MyDetails.class);
System.out.println(ob);
System.out.println(ob.getData().getClass().getName());
System.out.println(ob.getModels().getClass().getName());
System.out.println(ob.getDesign().getClass().getName());
System.out.println(ob.getMyinfo().getClass().getName());
}

}
```

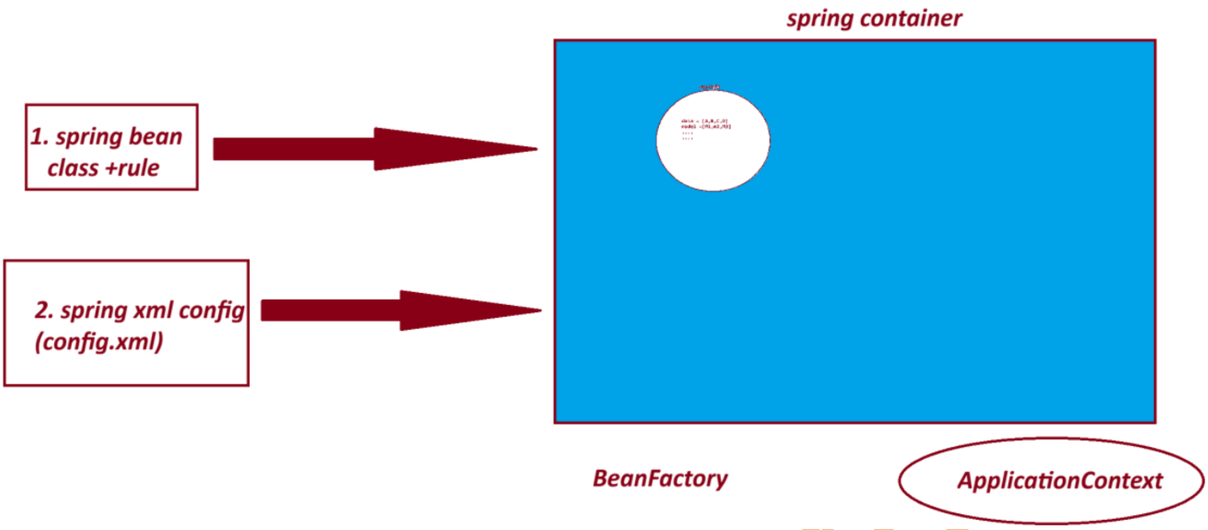
----

console output

```
MyDetails [data=[A, B, C, A], models=[M1, M2, M3], design={101=AB, 102=Suraj, 103=null}, myinfo={A1=B1, A2=B2, A3=B3, A5=}]
java.util.ArrayList
java.util.LinkedHashSet
```

java.util.LinkedHashMap

java.util.Properties



===

session 9